

VisualBubble: Exploring How Reflection-Oriented User Experiences Affect Users' Awareness of Their Exposure to Misinformation on Social Media

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ABSTRACT

Current solutions addressing misinformation on social media appear to rely on the misconception that misinformation is predominately spread by Artificial Intelligence (AI). However, the proliferation of false news is mainly due to humans. Solutions to curb misinformation should therefore emphasize human behavioral interventions, rather than focus solely on curbing AI bots. In this study, we analyze social media users' behaviors by means of a user journey. We create VisualBubble, a design probe that encourages reflection-oriented user experiences during news consumption on social media. We test our design probe with 10 users, to determine its effectiveness in increasing users' critical reflection on their news consumption behaviors. The initial findings show that VisualBubble can contribute to more critical attitudes towards the news that users are exposed to and, therefore, has the potential to mitigate social media misinformation.

CCS CONCEPTS

• Human-centered computing → Visualization design and evaluation methods.

KEYWORDS

Social media, misinformation, user experience, reflection, design, filter bubble

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1 INTRODUCTION

The emergence and widespread adoption of social media have changed the way news is formed and released. Consuming news through social media has become easier and cheaper [17]. However, this change also brings along the risk of misinformation. In 2020,

fake news on COVID-19 spread wildly, which led to social chaos and even deadly consequences [2]. In this research, the umbrella term "misinformation" refers to "all false or inaccurate information that is spread in social media" [22]. Researchers and social media platforms have put great effort into removing misinformation created by AI bots [5, 10, 15, 16, 19, 24]. However, studies found that false news spreading quicker, deeper, and broader than truth is mainly due to humans, rather than AI bots [20], although this human behavior is not always intentional. Two factors might contribute to the unintentional spread of misinformation by humans. First, people seem to be overconfident in their own judgments. A national survey conducted in the U.S. shows that 75.1% of the respondents are very confident that they can identify misinformation, but only 18.2% strongly agree that others can do it well [21]. Second, social media algorithms are designed to maximize users' exposure to contents that are congenial to them, but that are not necessarily reliable and true [25]. This phenomenon is called "filter bubble" and it refers to virtual spaces where users' beliefs are amplified by repetitive exposure to consistent opinions and contents. Filter bubbles can hinder users' critical thinking and their ability to detect false news. These facts show that solutions to curb misinformation should emphasize human behavioral interventions, rather than focus solely on AI bots [20].

In this late-breaking work, we first analyze and visualize social media users' news consumption behaviors through a user journey. Next, we describe a design probe, VisualBubble, aimed at increasing users' critical reflection on their news consumption. We report a study in which we test VisualBubble with 10 users, and we discuss the main results. The initial findings show that VisualBubble can contribute to more critical attitudes towards the news that users are exposed to and, therefore, has the potential to mitigate social media misinformation.

2 RELATED WORK

2.1 Transparency Tools

Transparency tools improve the understanding of algorithms' effects or allow comparing a user's digital space with those of other users. Their goal is to make people aware of why they are exposed to certain information and to add perspectives from outside their networks. Research shows that opinion and network polarization can contribute to the spread of misinformation [18]. Social media algorithms tend to reinforce this phenomenon, by showing users only what they like to read and by gradually forming polarized



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“filter bubbles” [18]. Although social scientists argue that filter bubbles would generally not turn as polarized as people fears [4], the “black box” property [13] of social media algorithms makes it hard for users to know if and how much their news exposure and consequent opinions are polarized. There are various ways to make this “black box” more transparent. Gobo, a web platform, tries to increase transparency by explaining how algorithms work and by enabling users to regain control over their news feeds, by managing and filtering contents [1]. Another example is a filter bubble visualization design from Nagulendra & Vassileva [9]. This design allows users to see which of their friends’ feeds they are currently exposed to, which visually explains how algorithms manage friends’ influence on social media. A web extension called Bobble helps users compare their results on search engines with the results shown to other users who input the same query [23], which is another way to increase transparency by comparison.

2.2 Reflection Tools

Reflection tools help users to reflect on their news consumption behavior and foster critical thinking. By encouraging rapid scroll, social media aim to increase users’ engagement with loads of news, which may discourage people from reflecting on the accuracy of the information [11]. Reliable news is usually mixed with many emotionally engaging news posts targeted to users’ interests, where accuracy is not always relevant. In this context, users may see social media as a place to relax and have fun, rather than a place to think critically [11], which might be one of the reasons for the quick spread of misinformation on these platforms. However, helping users reflect does not match the goal of social media platforms, since reflection takes time and might slow down consumption, thereby decreasing emotional engagement. As a consequence, social media platforms such as Facebook and Instagram do not show browsing history. Others, like YouTube, provide watching history, but do not analyze it in a way that helps users to reflect on their behavior. Some services were designed to help users reflect on what they have consumed. One browser extension called Scoopinon classifies the consumed information according to its source [12]. Another one, Balancer, helps users reflect on their own biases [8].

These design interventions encourage people to reflect on different aspects of their news consumption behavior, but more perspectives could be added. For example, helping users reflect on their news feeds by categorizing them into facts, opinions, or jokes might increase their awareness of what types of information they consume, eventually affecting their attitude in approaching social media. However, more work is needed to study how to keep a balance between reflection time and consumption engagement.

3 DESIGN PROCESS

3.1 User Journey

The design interventions mentioned above address specific phases of the user’s experience on social media. Gobo can be used when consuming news posts, while Scoopinon and Balancer are helpful after completing the news consumption. There might be more phases that deserve to be explored and addressed. Therefore, in the first step of our research, we set out to identify all phases of the news consumption experience on social media, and to visualize

them through a typical user journey, which can be used as a basis for generating new design interventions. Six participants (age: 20-65) who identified themselves as regular consumers of news on social media platforms participated in semi-structured interviews designed to explore their experiences of consuming social media news feeds. As a result, a user journey was generated which defines essential user experience phases (Figure 1).

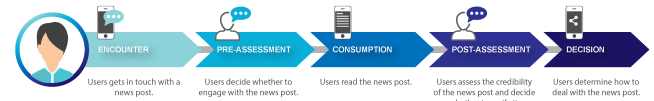


Figure 1: The user journey of consuming a news post on social media.

3.2 Design Workshop

With the insights from the user journey, a design workshop was performed with eight experts in either design or journalism. Design concepts aimed at reducing the spread of misinformation were generated for each phase of the user journey. Concepts were rated according to their feasibility and effectiveness. One design concept, VisualBubble, was selected to be prototyped and tested with users. The user tests were aimed at assessing the effectiveness of the design probe in increasing users’ critical attitude on their news consumption, compared to an existing social media platform.

4 DESIGN PROBE: VISUALBUBBLE

VisualBubble was designed for the pre-assessment phase of the news consumption journey, and was inspired by the ‘filter bubble’ phenomenon. It aims to ‘nudge and boost’ users to have increased resistance to misinformation [6] by showing them the existence of filter bubbles and even by exposing them to news outside such bubbles. The goal of this design is to encourage users to reflect on their current news consumption behavior.

Three filters are used to represent the bubbles the user is trapped in: Topic filter, Opinion filter, and Source filter. Each filter corresponds to one criterion used by AI filtering algorithms. The Topic filter (Figure 2) categorizes a user’s news feed by topic. Each news post is associated with one topic and visualized in the corresponding bubble. All topics are visualized as bubbles containing groups of posts. Users can zoom in on the bubbles to read the news. Outside “Your bubble” are the topics that the user is not exposed to but that are popular in the region. The Opinion filter (Figure 3) is based on the sentiment value of news, i.e. positive, negative, or neutral [7]. This filter first categorizes the news by topic and then by sentiment, creating sub-bubbles for each sentiment within a topic. The sub-bubbles with a black outline include news posts that mainly reflect negative sentiments towards the topic, while the white ones represent positive sentiments. “Opposite bubbles” on the right show news posts holding opposite opinions compared to “Your bubble” on the left. When the user zooms in on one side, the other side of the bubble also zooms in simultaneously. This way, the user can always easily see the corresponding opposite opinion on a certain topic. Source filter (Figure 4) categorizes a user’s feed by source.

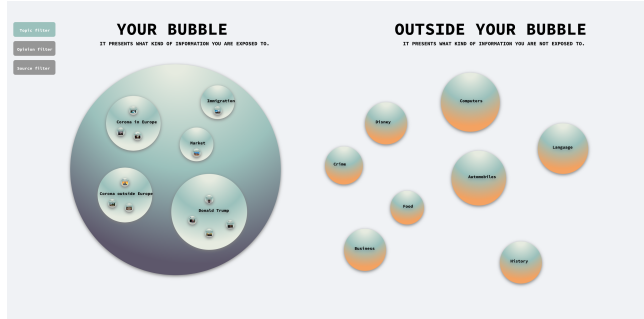


Figure 2: VisualBubble interface (Topic filter)

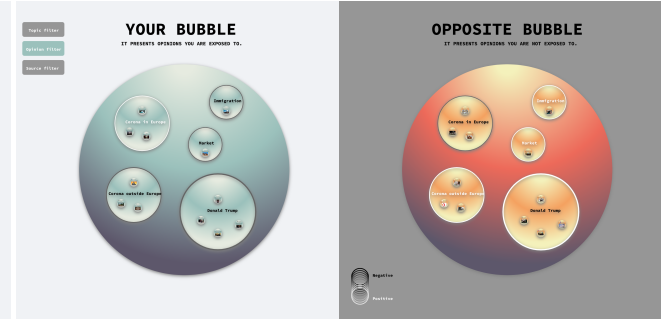


Figure 3: VisualBubble interface (Opinion filter)

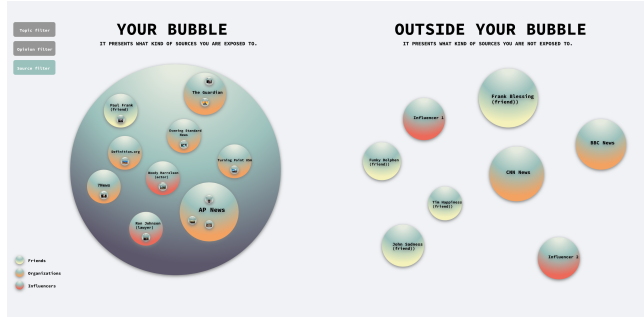


Figure 4: VisualBubble interface (Source filter)

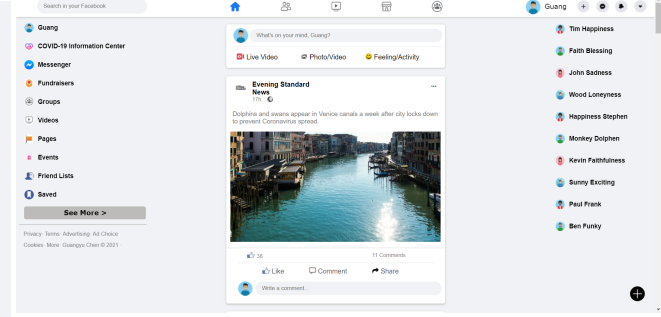


Figure 5: Facebook simulation interface

Unlike Nagulendra & Vassileva's filter bubble visualization design [9], VisualBubble not only shows friends as influencing sources, but also other sources such as social media influencers or organizations the user is following. Outside "Your bubble" are the sources that the user is not exposed to but that are popular in the region.

Compared to other existing tools, VisualBubble uncovers more aspects of the algorithms by offering the filter choices and, therefore, more space for reflection. Besides, it provides an alternative way to browse news feeds rather than infinite scrolling. It is worth mentioning that this design is not meant to replace the traditional way of consuming news feeds, but to provide an additional mode the users can switch to when they want to reflect on their news consumption.

5 USER STUDY

5.1 User Study Setup

VisualBubble is aimed at encouraging critical reflection by providing more perspectives and references, rather than by directly telling users whether a news post contains misinformation. In this way, users are encouraged to reflect on their social media feeds themselves which, in turn, might increase their critical attitude towards the news they consume. To explore how reflection-oriented user experiences may affect users' awareness of their exposure to social media misinformation, we performed a user study on VisualBubble. In this study, we analyzed the effects of VisualBubble compared to a traditional social media platform.

10 participants (age range: 20-30) were involved in the study. None of the participants was previously involved in the user journey semi-structured interviews. They all used at least four social media platforms on a regular basis, and had experience consuming news on Facebook. To assess how users perceive misinformation in VisualBubble compared to existing social media platforms, we

used a simulated Facebook interface (Figure 5) as the baseline of our test. Both VisualBubble and the Facebook simulation were built in the form of web pages. 12 news-related posts from 2020 and 2021 were selected and presented on both web pages. The posts contained 5 "True", 4 "False", and 3 "Not accurate" news. "True" means that all elements from the post are true. "False" means that all elements from the post are false. "Not accurate" means that the post has significant elements of both truth and falsity. All news was fact-checked.

5.2 Procedure

During a 50-minute interview, each participant was asked to interact with both interfaces and answer some questions. Each interview included two tests. Test 0 was the baseline test to record how the participant interacts with and perceives the traditional social media (Facebook) interface. The participant was first asked to freely browse the Facebook-like interface and then to rate their perceived exposure to different topics, opinions, and sources through a Likert scale (1=strongly disagree; 3=neutral; 5=strongly agree) (see Table 2). After that, they were asked to assess the accuracy of the 12 news posts by assigning one of the following labels: "Cannot assess", "True", "False", or "Not accurate". Following Test 0, Test 1 was performed. Test 1 was aimed at assessing the effectiveness of VisualBubble. The same 12 news posts presented in Test 0 were categorized into different bubbles, according to the three filters: Topic, Opinion, and Source. The participant was asked to browse the news feeds in VisualBubble and to freely interact with them for 5 minutes. Then they were asked to fill in the same Likert scales from Test 0. After that, the participant had to assess each post's accuracy again. In both tests, they were not informed about the correctness of their assessments. At the end of Test 1, the participant was asked about their opinions on VisualBubble and whether it

could contribute to encouraging critical thinking and to mitigating misinformation on social media (see Figure 6).

All the interviews were conducted in Microsoft Teams due to COVID-19 restrictions. The study was approved by the university's Ethical Review Board. The audio and the screen from the participants were recorded, and then transcribed and converted into written notes for subsequent analysis.

6 RESULTS

6.1 Increased assessments and skepticism

In total, 120 news posts were assessed (12 news posts, 10 participants) for both Test 0 and Test 1. Results are reported in Table 1. By comparing the assessments of news posts in Test 0 and 1, several observations emerged. First, news posts that were labeled by participants as "Cannot assess" decreased from 27% to 22% in Test 1. This shows that participants became more confident in making assessments after using VisualBubble. Second, the correct assessment of misinformation ("Not accurate" news posts) increased from 44% to 51%, which indicates that participants became better at detecting misinformation, by being exposed to opposite opinions or by reflecting on their exposure to news topics. However, the correct assessment of "True" news decreased from 93% to 88% showing that, after using VisualBubble, participants became over-skeptical. In the end, the overall correct assessment stayed exactly the same (68%). To summarize, by using VisualBubble as an "accuracy prompt" [3], participants became more willing to make assessments and more critical, but they also showed the tendency to become over-skeptical.

6.2 Topics exposure

Results from the Likert scale in Test 0 and 1 were analysed through the paired sample one tail T-test. The Cohen's effect size was calculated as well. When comparing the agreement level with the statement *"I am aware of the topics I am being exposed to"*, users reported a statistically higher level of awareness after using VisualBubble (Table 2). During test 1, one participant commented, *"(with VisualBubble), I can easily know how the system promotes news to me by topic."* Another participant stated, *"With the linear way to scroll feeds, I wouldn't think about the topics of my feeds... But VisualBubble gives me a holistic view of them"*. These results suggest that the Topic filter made it easier for users to be aware of the topics they are exposed to.

When asked to comment on this filter, some issues emerged. One participant reported that the algorithm that categorizes topics might overlook certain topics included in a news item, or may subjectively interpret their hierarchy. This participant decreased his agreement level on this statement in test 1 and commented, *"(topic filter) might narrow down the topics I'm actually consuming... Each post normally links to several topics. I don't know why it categorizes in this way..."* This concern is justified. For example, the news *"Dolphins and Swans Appear in Venice Canals a Week After City Locks Down to Prevent Coronavirus Spread"* is related to topics such as "Covid-19", "environment", "sea life", etc. Putting this news just under the "Covid-19" topic emphasizes its relevance to "Covid-19", but may distract people from its relevance to the "environment". However, placing the same piece of news under more than one

topic bubble would lead to information redundancy. Therefore, an objective hierarchical mechanism might be needed to rank the relevance of topics covered in each news post, although it would be difficult to create. Alternatively, more complex visualizations might be needed, with the risk of making the interface less clear and effective. Nevertheless, participants' criticism on this aspect might bear some positive aspects, because with the traditional linear way to scroll feeds many participants would not even think about what topics they consume. The Topic filter at least evoked questions and sparked reflections and critical thinking, which are all very important elements in mitigating the spread of misinformation [14].

6.3 Opinion exposure

When comparing users' agreement with the statement *"I am aware of the opinions I am being exposed to"*, test 1 did not have a statistically significant increase over test 0 (Table 2). The main reason could be that participants doubted the principle of categorizing news into positive or negative opinions within certain topics. They pointed out that, if moved to another topic, the same news might change valence. For example, news feed such as *"the advent of 5G"* might be positive if categorized under the "technology" topic, but negative if categorized under "privacy". The subjectivity aspect also made some participants worry about manipulation and led them not to trust the opinions they were exposed to. Besides, the Opinion filter might be even more subjective than the Topic filter. *"There are a thousand Hamlets in a thousand people's eyes"*: different people can interpret the same piece of news differently, and it's likely that these interpretations are incompatible, unlike the Topic filter, where different categorizations can be compatible. Therefore, designing an Opinion filter and getting people to understand and trust it can be very challenging. One possible solution might be highlighting alternative opinion(s) with neutrally colored outlines - instead of strictly labeling them as 'opposite' opinions.

Although participants did not agree much on the classification of the opinions they were exposed to, they felt exposed enough to different opinions through the "Opposite opinion bubbles". When comparing the agreement level with the statement *"I feel exposed enough to different opinions"*, test 1 had a statistically significant increase over test 0. (Table 2). About the "Opposite opinion bubbles", one participant commented: *"I think it helps me reflect on the opinions I already had and maybe I will accept new ideas..."*. Most participants agreed that the "Opposite opinion bubble" can help them expand their opinions and thus be less polarized. Next to that, a design problem was mentioned by some participants during test 1. They did not easily perceive the difference between the opinions included in their bubble and the ones belonging to the opposite bubble and needed some explanation by the researcher. The Opinion filter should therefore include news of opposing opinions where the opposite is more obvious. However, it is unlikely to find two completely opposite news items in real life. In order to give users a better understanding of the Opinion filtering effect, the opposites might need to be explained more clearly in the interface.

6.4 Source exposure

In Table 2, both statements related to the Source filter have large effect size and are statistically significant. Participants agreed that

	Test 0	Test 1	Annotation
News that have been declined to assess	27%	22%	In Test 1, participants were more willing to make assessments.
Correct assessment of misinformation ("False" and "Not accurate" news)	44%	51%	In Test 1, participants were better at assessing misinformation.
Correct assessment of "False" news	30%	38%	In Test 1, participants were better at assessing False news.
Correct assessment of "Not accurate" news	59%	63%	In Test 1, participants were better at assessing inaccurate news.
Correct assessment of "True" news	93%	88%	In Test 1, participants were worse at assessing True news.
Overall correct assessment	68%	68%	In Test 1, the overall correct assessment did not change.

Table 1: Comparison between test 0 and test 1 on the judgment results of 12 news.

Statement	Test	Mean (1-Strongly disagree; 3-Neutral; 5-Strongly agree)	Effect size	T-value	P-value
Topic filter					
I am aware of the topics I am being exposed to.	Test 0	4.1	0.98: Large	2.71	0.01
	Test 1	4.7			
I feel exposed enough to different topics.	Test 0	3.4	0.98: Large	1.63	0.06
	Test 1	4.2			
Opinion filter					
I am aware of the opinions I am being exposed to.	Test 0	4.3	0.00: Negligible	0.00	0.50
	Test 1	4.3			
I feel exposed enough to different opinions.	Test 0	3.3	0.79: Medium	2.69	0.01
	Test 1	4.0			
Source filter					
I am aware of the sources I am being exposed to.	Test 0	4.2	1.11: Large	2.99	0.01
	Test 1	4.7			
I feel exposed enough to different sources.	Test 0	3.0	1.01: Large	2.99	0.01
	Test 1	4.0			

Table 2: T-test results of the Topic, Opinion and Source filters' perception between Test 0 and Test 1.

the Source filter could help them be aware of the sources they were exposed to, and the news in "Outside your bubbles" made them feel more exposed to different sources. The Source filter was the only filter that did not raise any doubts on its categorization mechanism. This may likely be due to its higher perceived objectivity. Indeed, people and organizations who post news are registered on the social media platform, and can be easily identified, therefore participants seemed to trust this filter more, compared to the others. Although the identity of social media users cannot always be verified, and therefore it bears some level of uncertainty and risk, this point was not raised by participants.

6.5 Mitigating social media misinformation

Almost all participants claimed that VisualBubble could not mitigate misinformation directly but in an indirect way. They agreed that VisualBubble could help them be more critical towards the news they are exposed (Figure 6a) to and could potentially limit social media misinformation (Figure 6b). One participant said: *"With more perspectives shown to me, I will more carefully consider if a feed is true or not..."* Another one commented: *"Different overviews of my feeds let me reflect how my current consumption behavior is... I might be more conscious how reading new contents affects my overall performance..."* Another participant claimed that *"The information outside of my bubble can kind of help to reference if I'm stuck in some extreme opinions..."* One participant even utilized the Source filter as a new way to assess news: when they assessed as reliable most news posts from a source, they would also trust the other news posts from the same source. In general, these filters emphasized different

perspectives on news posts and provided users with new viewpoints outside their own networks, which stimulated critical thinking and created reflection moments. Therefore, the filters turned out to be potentially effective in reducing misinformation.

6.6 New consumption habits

9 out of 10 participants agreed or strongly agreed that they would use this design if it was available on the market in the future (Figure 6f). They thought that VisualBubble would change their consuming habits (Figure 6c) and could make the consuming process slower (Figure 6d). Most participants stated that they value accuracy more than efficiency, and that they would not mind spending more time to ensure the accuracy of the news they consume (Figure 6e).

7 DISCUSSION

7.1 The need for transparency in filtering algorithms

With the traditional way of browsing news feeds, the mechanisms and effects of filtering algorithms are not obvious. One participant noted that while using the linear feed, her attention was on the individual news posts, and there was an urge to scroll for the next ones. After a while, she would feel lost in loads of seemingly unstructured information and become slightly numb. In Test 1, with certain criteria of filtering algorithms being made evident, participants could start to identify and criticize some aspects of the filtering algorithm behaviors, and some of their decisions. By questioning the rules of the algorithm, participants started to become more critical of their

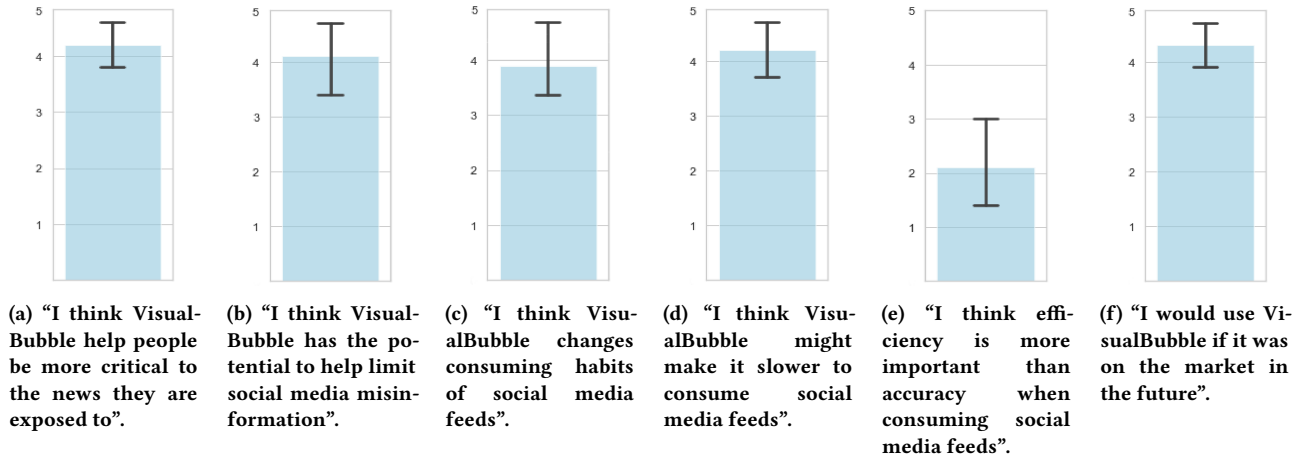


Figure 6: Agreement score on the statements (1=strongly disagree; 3=neutral; 5= strongly agree).

outcomes - the individual news pieces they were exposed to. This causal relationship was not validated in this study, but it should be explored in future research.

7.2 Subjective filters and trust issues

The use of subjective filters such as Topic and Opinion filters need to be further discussed and tested. People seemed to have positive feelings towards objective filters (such as the Source filter) but had more negative attitudes towards subjective filters (such as the Topic and the Opinion filters). Certain levels of bias might always accompany subjective filters, which will lead to trust issues or manipulation concerns. To solve this, some directions can be explored in future work, such as explaining the criteria underlying subjective filters, which could help users trust the system more, or giving users power over the current filtering algorithms. However, controlling what perspectives to see might lead to another level of polarization, one that is determined by the user. Therefore, some levels of exposure to different opinions might be beneficial to maintain. Lastly, it might be interesting to utilize manipulation concerns to increase the level of alertness in the user. Being aware that some categorizations are subjective might stimulate users' critical thinking and therefore make them more alert about potential misinformation.

7.3 Limitations

The initial findings reported in this paper are based on user tests with 10 participants. A larger sample of subjects, encompassing different age ranges, would help to validate these preliminary results in the future. During the user study, it is possible that participants could reflect on the accuracy of the news posts between Test 0 and Test 1. Therefore, the effect of seeing the same news post multiple times on users' assessment should be addressed in future research. Finally, the influence of the UI elements was not considered in this study. For example, the visual details of the interface, the relative size of each bubble, and the mouse wheel-controlled interaction might influence participants' understanding of the design probe. Assessing the influence of UI elements on users' interactions and reflections will help to optimize the usability and effectiveness of the solution.

8 CONCLUSION

This paper explores new ways to reduce the spread of misinformation on social media through a design-driven and experience-centered approach. The first contribution of this work is the user journey visualizing the process of information consumption on social media, which can serve as a basis for future design explorations addressing different phases of the news consumption process. The second contribution consists of the description and validation of the design probe, which aims at increasing users' awareness on the effects of filtering algorithms and their news consumption behaviors. By providing perspectives or reference elements that allow users to reflect on the news they consume, VisualBubble contributes to more critical and reflective attitudes towards news consumption behavior and, therefore, have the potential to mitigate the spread of misinformation on social media. In this preliminary study, the probe turned out to be effective in increasing users' willingness to assess the posts' accuracy and users' ability to correctly label misinformation, and users showed interest in its adoption.

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